

Mann Santosh Wani

B.Tech in Electronics & Telecommunication Engineering - Third year

CGPA: **9.60**

✉ mannwani.876@gmail.com

☎ +91-9730395810

🌐 [linkedin.com/in/mann-s-wani](https://www.linkedin.com/in/mann-s-wani)

🐙 github.com/Mann-Wani

Education

COEP Technological University, Pune (COEP) – B.Tech in Electronics and Telecommunication Engineering, CGPA: 9.60	2024 – 2028
Smt. Chandibai Himathmal Mansukhani College, Ulhasnagar, Thane – Higher Secondary Certificate (HSC) – Aggregate: 97.67% PCM: 99% JEE Advanced : AIR 12,550 MHT-CET PCM : 99.87 %ile	2022 – 2024
Carmel Convent High School, Badlapur, Thane – Secondary School Certificate (SSC): 96.4%	2010 – 2022

Proficiency & Skills

- **Programming** : C, C++, Java, Python, BASH, Assembly
- **Software**: Docker (Containers, Volumes, Ports, Multi-stage Build, Dockerfile, Docker Compose, Cgroups), Keil C51, STM32CubeIDE, ESP-IDF, Eclipse Mosquitto, Arduino IDE
- **Computer Systems** : Networking (TCP/IP, UDP, DNS, ARP, DHCP, NAT, MAC, OSI), Cloud & Servers, DSA (Linked Lists, Queues, Hash tables, Graphs, Recursion, Backtracking, Trees, Binary Search Trees), DBMS, OOP (Encapsulation, Inheritance, Multi-Threading, Method-overloading & overriding, Stack, Heap, Interfaces), Git
- **Operating Systems**: CPU scheduling, Processes & Threads, Bootloader, RTOS, File systems, Disk & Memory Management, Linux (Ubuntu)
- **Microcontrollers**: Raspberry Pi, Nvidia Jetson, ESP, STM, Arduino, 8051
- **Peripherals**: SPI, I²C, UART, I²S, Bluetooth, Wi-Fi, ESP-NOW, Timers, ADC, Interrupts, DMA, WDT, USB
- **Embedded Systems**: Sensor Interfacing, PCB Designing, Firmware development, Bare-metal programming, Embedded C, HAL, CMSIS, Toolchain, ISR, ISA, JTAG, SWD
- **Extra-Curriculars** : Member of COEP Satellite Team, Class Representative, Sports

Projects

- **CPU Scheduler Simulator** -
Developing a CPU Scheduling Simulator in C implementing FCFS, SJF, Priority & Round Robin algorithms. Simulated process execution using queues, generated Gantt charts & scheduling metrics such as waiting time, turnaround time & completion time.
- **AI Powered Voice Assistant** -
Developed a server based voice assistant which implements ASR (Whisper), LLM (Mistral), TTS (Piper) & Face recognition (InsightFace) AI models to actively hear to voice commands and respond to the queries. Interfaced RTC, OLED, Audio DAC (MAX98357A), Microphone (INMP441) along with ESP microcontroller. The software stack included use of TCP, PCM audio frames and RTOS for task scheduling. The server was deployed on the Linux kernel.
- **Control System for a 1-DOF UAV : TowerCopter** -
Served as the primary embedded firmware developer inorder to develop firmware for STM32F103C8T6, VL53L0X (TOF Sensor) & ESC for BLDC motors. Currently using Timers, PWM, ADCs to implement the project. The project aims to create a optimised control system for UAVs to tackle the IGE & OGE effects thus stabilising flight controls.

Work Experience

- **Bajaj Electricals R&D Centre, Navi Mumbai – IoT Intern** May 2026 – July 2026
Hands-on experience with Raspberry Pi, Nvidia Jetson, Eclipse Mosquitto, MQTT explorer, Ubuntu, Docker, LLMs, Networking, Cloud, Server
 - **Smart Street Lighting** -
Developed a smart street lighting solution involving use of C++ framework. Implemented MQTT protocol using Eclipse Mosquitto. Interfaced ESP Microcontrollers & Arduino IDE. Created a Proof-of-Concept using 1 Gateway and 2 Nodes. Developed the modified protocol stack to achieve reliable message relaying from gateway to end node. The stack also fetched health data from each node thus providing Individual Light Monitoring
 - **Offline Voice Controlled Switches** -
Designed a PCB using KiCAD and trained an Offline Voice Recognition module to control home appliances with voice commands without need of cloud connectivity or servers maintaining complete privacy. Implementation of 10 different voice commands having ability to control upto 4 devices

Awards & Achievements

- 1st Rank in ENTC Department, COEP for two consecutive years.
- Certificate of Merit from COEP Alumni Association and Frank Hirschvogel Components
- 1st Rank in 12th Boards HSC – Mumbai Region
- Recipient of Government Scholarship for excellence in HSC (Selected from entire Mumbai Region)